Trader Performance vs Market Sentiment

Objective

The objective of this project is to analyze how market sentiment (Fear and Greed) influences trader behavior and performance. By combining sentiment data with historical trading activity, the study aims to uncover patterns that can help improve trading strategies.

Methodology

Two datasets were used: a market sentiment dataset (Fear/Greed Index) and a historical trader dataset. Data preprocessing involved handling missing values, removing duplicates, and converting timestamps into a common date format. Both datasets were merged on a daily level to align trading activity with market sentiment.

Key metrics were created, including daily profit and loss (PnL), win rate, average trade size, leverage usage, number of trades per day, and long/short ratio. These metrics were analyzed across different sentiment categories such as Extreme Fear, Fear, Neutral, Greed, and Extreme Greed.

Key Insights

1. **Higher Profitability in Greed Markets:**
   Traders achieve the highest average profits during Extreme Greed conditions, indicating strong market confidence and aggressive trading behavior.
2. **Reduced Performance in Fear Conditions:**
   Profitability declines during Fear and Extreme Fear periods, reflecting uncertainty and cautious trading decisions.
3. **Increased Trading Activity in Greed:**
   The number of trades is significantly higher during Greed phases, suggesting that traders are more active in positive market conditions.
4. **Leverage-Driven Risk Behavior:**
   Traders tend to use higher leverage during Greed, which increases both potential returns and the risk of losses. High leverage traders show more volatile performance.

Strategy Recommendations

1. **Risk Reduction in Fear Markets:**
   During Fear and Extreme Fear conditions, traders should reduce leverage and limit trading frequency to minimize potential losses.
2. **Controlled Aggression in Greed Markets:**
   During Greed and Extreme Greed conditions, traders can increase participation but should apply strict risk management strategies such as stop-loss to control downside risk.

Conclusion

The analysis demonstrates that market sentiment plays a crucial role in shaping trader behavior and performance. Traders tend to be more aggressive and profitable during Greed phases, while Fear leads to cautious and less effective trading. Incorporating sentiment-based adjustments into trading strategies can help improve decision-making and overall performance.